

Debate Generation System in Japanese Rap Battle Format

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1 Abstract

We propose a "Debate System in Japanese Rap Battle Format." Our goal is to elevate discussions into entertainment, encouraging people of all generations to form their own opinions.

This system generates two distinct opinions (lyrics) when presented with a discussion topic, and based on these opinions, agents engage in a debate in the form of a rap battle. The system's technical composition incorporates several innovative elements:

- (1) Lyric (Opinion) Generation: The system creates lyrics that represent opinions on the given topic. A unique feature of this process is its consideration of rhymes specific to the Japanese language. This ensures that the generated content not only conveys meaningful opinions but also adheres to the rhythmic and phonetic patterns characteristic of Japanese rap.
- (2) Contextual Awareness: The system is designed to understand and maintain the context of the ongoing debate. This contextual awareness enables coherent and relevant exchanges



Figure 1: Debate Generation System in Japanese Rap Battle Format

between the agents, simulating the flow of a real debate or conversation.

- (3) Responsive Utterances: Building on its contextual awareness, the system generates responses that directly address the opponent's previous statements. This feature creates a dynamic, interactive debate environment, where each agent's contribution is influenced by and responds to the other's arguments.
- (4) Beat-Adaptive Audio Synthesis: To enhance the rap battle experience, the system incorporates audio synthesis that adapts to the underlying beat. This synchronization between the generated lyrics and the musical rhythm results in a more authentic and engaging rap performance.

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(5) Integration into Audio-Visual Performance: All these elements—generated lyrics, contextual responses, and beatsynchronized audio—are combined to create a comprehensive audio-visual performance. This integration transforms the debate from a mere exchange of words into an entertaining

and immersive experience. The culmination of these technical components results in a unique performance where complex debates are conducted through the medium of a Japanese rap battle, offering an innovative blend of technology, language, and music.